

Arithmetic Sequences as Linear Functions — Off-by-One Index Errors

Answer Key

Algebra 1

Quick Answers

1. 51	3. 17	5. 378	7. 784
2. 11	4. 15, 15	6. 136	8. —

Curriculum

Level: Algebra 1

HSF-BF.A.2, HSF-LE.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. If they got 56: multiplied the common difference by n instead of by n minus one, which takes one step too many
 2. If they got 7: read the constant in the linear rule as the first term, but that constant is the value at n equals zero
 3. If they got 16: subtracted the two index numbers and forgot that both endpoints are themselves terms of the list
 4. If they got 14: solved the equation a plus d times n equals the value, which counts one step too many and lands one term early
 5. If they got 408: used a plus twelve times the common difference as the last term, so the sum ran one term too far along
 6. If they got 143: used the first term as the constant of the linear rule instead of the value one step before it
 7. If they got 735: counted the terms by subtracting the two index numbers, giving one term fewer than the list actually holds
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1. Find and fix: Rosa's 10th term, $a_1 = 6$, $d = 5$.

Step 1: name the slip. The first term is already in hand before any step is taken, so reaching the 10th term takes 9 steps, not 10. Rosa multiplied d by n instead of by $n - 1$.

Step 2: use the explicit rule as written: $a_n = a_1 + (n - 1)d = 6 + (10 - 1)(5) = 6 + 45$.

Step 3: the 10th term is 51.

Step 4: check with the $n = 1$ test — the correct rule gives $6 + 0 \cdot 5 = 6$, the first term ✓; Rosa's version gives $6 + 5 = 11$ at $n = 1$, which is already the *second* term.

Watch for: 56, exactly one common difference too large.

2. Find and fix: Theo reads $a_n = 4n + 7$.

(a) Step 1: substitute the index the constant belongs to. Setting $n = 0$ gives $4(0) + 7 = 7$, so the constant is the value *one step before* the sequence starts — the “zeroth term”, not a_1 .

(b) Step 2: the first term is the rule evaluated at $n = 1$: $a_1 = 4(1) + 7$.

Step 3: $a_1 = \boxed{11}$.

Step 4: check the step size — $a_2 = 4(2) + 7 = 15$ and $15 - 11 = 4 = d$ ✓. Theo’s answer 7 sits exactly one common difference below the real first term, which is the fingerprint of this error.

Teaching point: in $a_n = dn + b$ the constant is always $a_1 - d$. On a graph it is the y -intercept, and the sequence’s own points start at $n = 1$, one step to the right of it.

3. Find and fix: how many terms from the 7th to the 23rd?

Step 1: name the slip. Subtracting index numbers counts the *gaps* between terms, not the terms. Going from 7 to 23 takes 16 steps, but each step lands on a new term *and* the starting term was already there.

Step 2: count with both endpoints included: $23 - 7 + 1 = 17$.

Step 3: the list holds $\boxed{17}$ terms.

Step 4: check on a list small enough to count by hand — from the 7th to the 9th term is 7, 8, 9, three terms, and $9 - 7 + 1 = 3$ ✓ while $9 - 7 = 2$ is wrong.

Watch for: 16. This “fencepost” slip is the same off-by-one as the $(n - 1)$ in the explicit rule, seen from the other side.

4. Find and fix: which term of 5, 11, 17, ... equals 89?

(a) Step 1: the student’s equation $5 + 6n = 89$ is missing the index adjustment. The explicit rule gives $a_n = 5 + 6(n - 1)$, so the equation to solve is $5 + 6(n - 1) = 89$.

(b) Step 2: solve it. $6(n - 1) = 84$, so $n - 1 = 14$.

Step 3: $n = \boxed{15}$ — the 15th term equals 89.

Step 4: substitute back: $5 + 6(15 - 1) = 5 + 84 = 89$ ✓. The student’s equation returns 14, and $5 + 6(14 - 1) = 83$, not 89.

Watch for: 14. Note that the wrong equation produced a perfectly clean whole number — an index error hides best exactly when it does, which is why substituting back is not optional.

5. Find and fix: Priya’s sum of twelve terms, $a_1 = 4$, $d = 5$.

Step 1: her formula $S_n = \frac{n}{2}(a_1 + a_n)$ is correct. The error is in a_n : she used $a_1 + nd = 4 + 60 = 64$, which is the *thirteenth* term, so her sum ran one term too far along.

Step 2: the true twelfth term is $a_{12} = 4 + (12 - 1)(5) = 4 + 55 = 59$.

Step 3: $S_{12} = \frac{12}{2}(4 + 59) = 6 \times 63 = \boxed{378}$.

Step 4: check with the other form, $S_n = \frac{n}{2}(2a_1 + (n - 1)d) = 6(8 + 55) = 378$ ✓.

Watch for: 408. The gap of 30 is exactly six common differences — half of the twelve terms — the signature of a last term that is one step too far.

6. Find and fix: Dana writes 3, 10, 17, 24, ... as $a_n = 7n + 3$.

(a) Step 1: run the $n = 1$ test on Dana's rule: $7(1) + 3 = 10$. That is a_2 , not a_1 , so the rule is shifted one place along the sequence.

(b) Step 2: build the correct linear form. The slope is the common difference, $d = 7$, and the constant is the value at $n = 0$, one step below the first term: $b = a_1 - d = 3 - 7 = -4$. So $a_n = 7n - 4$, and the $n = 1$ test now gives $7 - 4 = 3$ ✓.

Step 3: $a_{20} = 7(20) - 4 = 140 - 4 = \boxed{136}$.

Step 4: cross-check with the explicit rule: $a_{20} = 3 + (20 - 1)(7) = 3 + 133 = 136$ ✓.

Watch for: 143, which is a_{21} . Using a_1 as the constant always returns the term one place further on.

7. Find and fix: the sum of a_5 through a_{20} for $a_n = 3 + 4(n - 1)$.

(a) Step 1: count the terms with both endpoints included: $20 - 5 + 1 = 16$ terms. The classmate used $20 - 5 = 15$, counting gaps rather than terms.

Step 2: the two end terms are $a_5 = 3 + 4(4) = 19$ and $a_{20} = 3 + 4(19) = 79$; both were computed correctly, so the count is the only error.

(b) Step 3: $S = \frac{16}{2}(19 + 79) = 8 \times 98 = \boxed{784}$.

Step 4: check by subtracting whole sums — the first 20 terms total $\frac{20}{2}(3 + 79) = 820$ and the first 4 total $\frac{4}{2}(3 + 15) = 36$, and $820 - 36 = 784$ ✓.

Watch for: 735. The shortfall of 49 is exactly the average term, so the wrong answer looks like a plausible sum — only the count betrays it.

8. Explain: steps, the zeroth term, and why the error hides.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.

(a) The sequence is already sitting on a_1 before anything happens, so the first term costs zero steps. Every later term costs one common difference per step, and from term 1 to term n there are $n - 1$ steps, so the rule adds $(n - 1)d$. Multiplying by n pays for one step that was never taken, which is why the answer is always exactly one d too large.

(b) Writing $a_n = dn + b$ treats the index as an input to a line. The constant b is the output at $n = 0$, but the sequence has no zeroth term — it starts at $n = 1$. Since one step of the line is worth d , going back from a_1 to $n = 0$ subtracts one: $b = a_1 - d$. The line is a continuous object that happens to pass through the sequence's points; the sequence only occupies the integer inputs from 1 onward, so the intercept belongs to the line, not to the list.

(c) Two checks. (i) *The $n = 1$ test:* substitute $n = 1$ into any rule you write; it must return the first term, which exposes nd written for $(n - 1)d$ in a single line. (ii) *Substitute the answer back:* put the term number you found into the original sequence and confirm it produces the value the problem asked about. A third habit worth credit: on a counting question, test the counting method on a tiny list you can check by hand, e.g. terms 7 through 9.

What a full-credit answer must contain: the first term costing no steps (or an equivalent count of the gaps) in (a), the constant identified as the value at $n = 0$, i.e. $a_1 - d$, in (b), and two checks in (c) that are genuine *tests* rather than restatements of the formula.